

MATH

21/11/25

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EX-a B

3) 45 cows can graze a field in 13 days. How many cow will graze the same field in 9 days?

No. of cow req. to graze in 13 days = 45 cows

No. of cows req. to graze in 1 day
 $= (45 \times 13) \text{ cows}$

$\therefore$ No. of cows req. to graze in 9 days.

$$= \left(\frac{45 \times 13}{9} \right) \text{ cow} = 65 \text{ cows}$$

4) 16 horses can consume a certain quantity of corn in 25 days. In how many days would the same quantity be consumed by 40 horses?

16 horses can consume corn in = 25 days
 1 horse can consume corn in = (25×16) days

40 horses can consume the corn in

$$= \frac{25 \times 16 \times 4}{40 \times 12 \times 3}$$

$$= \left(\frac{25 \times 16^2}{40 \times 8} \right) \text{ days} = 10 \text{ days}$$

5) A girl can finish a book in 25 days. If she reads 18 pages in a day. How many days will she take to finish it if she reads 15 pages everyday?

If she reads 18 pages everyday, then the book will finish in = 25 days.

If she reads 1 page everyday, then the book will finish in = (25 × 18) days

∴ If she reads 15 pages everyday, then the book will finish in

$$= \left(\frac{25 \times 18}{15} \right) \text{ days} = 30 \text{ days}$$

6) Reeta types 40 words per minute and takes 24 mins to type a certain document. Her friend Leta types 48 words per minute. In how much time, will she be able to type the same document?

In 40 words per minute, it takes 24 mins to finish a document.

So,

In 1 words per minute it takes
 $= (24 \times 40)$ mins.

∴ In 48 words per min it takes

$$= \left(\frac{24 \times 40}{48} \right) \text{ mins} = 20 \text{ mins}$$

- 1) A bus covers a certain dist. in 3 hrs 20 mins at an average speed of 45 km/hr. How long will it take to cover the same distance at a speed of 36 km/hr

In 45 km/hr, it takes to cover = 3.20

In 1 km/hr, it takes to cover = (3.2×45) hrs

∴ In 36 km/hr, it takes to cover

$$= \left(\frac{32 \times 45}{36} \right) = 4 \text{ hrs}$$

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- 8) At the beginning of a month a factory has enough materials to make 240 tons of steel in a month. If 60 more tons of steel is to be made. How long will the materials last?

GEOGRAPHY

Climate of India

Tropical Monsoon Climate →

India lies between $8^{\circ}4'N$ to $37^{\circ}6'N$ latitudes with tropic of cancer passing through its middle giving India its tropical climate.

It experiences seasonal change in wind direction and therefore is named to have a monsoon climate which means season thus India has a tropical monsoon climate.

- 'India is a country of great climatic diversity' Justify.

- Area is having a warm tropical climate with heavy rainfall for most part of year.
eg- North East India, Andaman and Nicobar Island, Kerala coast.

Tropical monsoon climate

India is having a tropical monsoon climate getting seasonal rainfall like the Central India Deccan Plateau & the northern plains.

Hot desert climate

- Very dry climate with scanty rainfall. This area is sandy desert. eg Rajasthan.

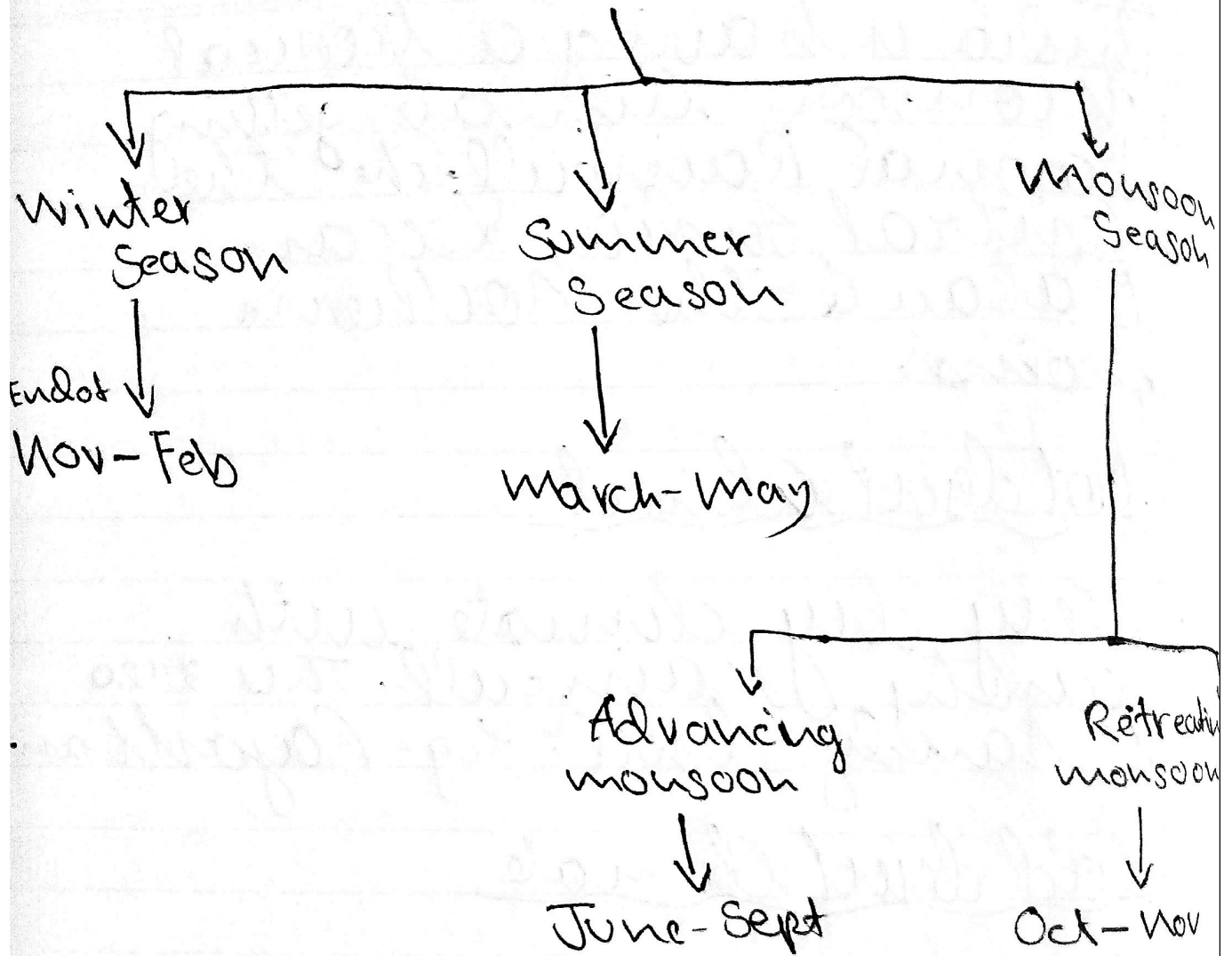
Cold desert climate

- It has extremely cold winter and cold summer. The region is snow covered for most part of year.
 eg - The cold deserts of Ladakh.

Cool temperate climate

- This region has cold winters and pleasantly warm summer.
 eg - Hill stations of the Himalayas.

Seasons of India



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like Shimla and Darjeeling.

Seasons of
India

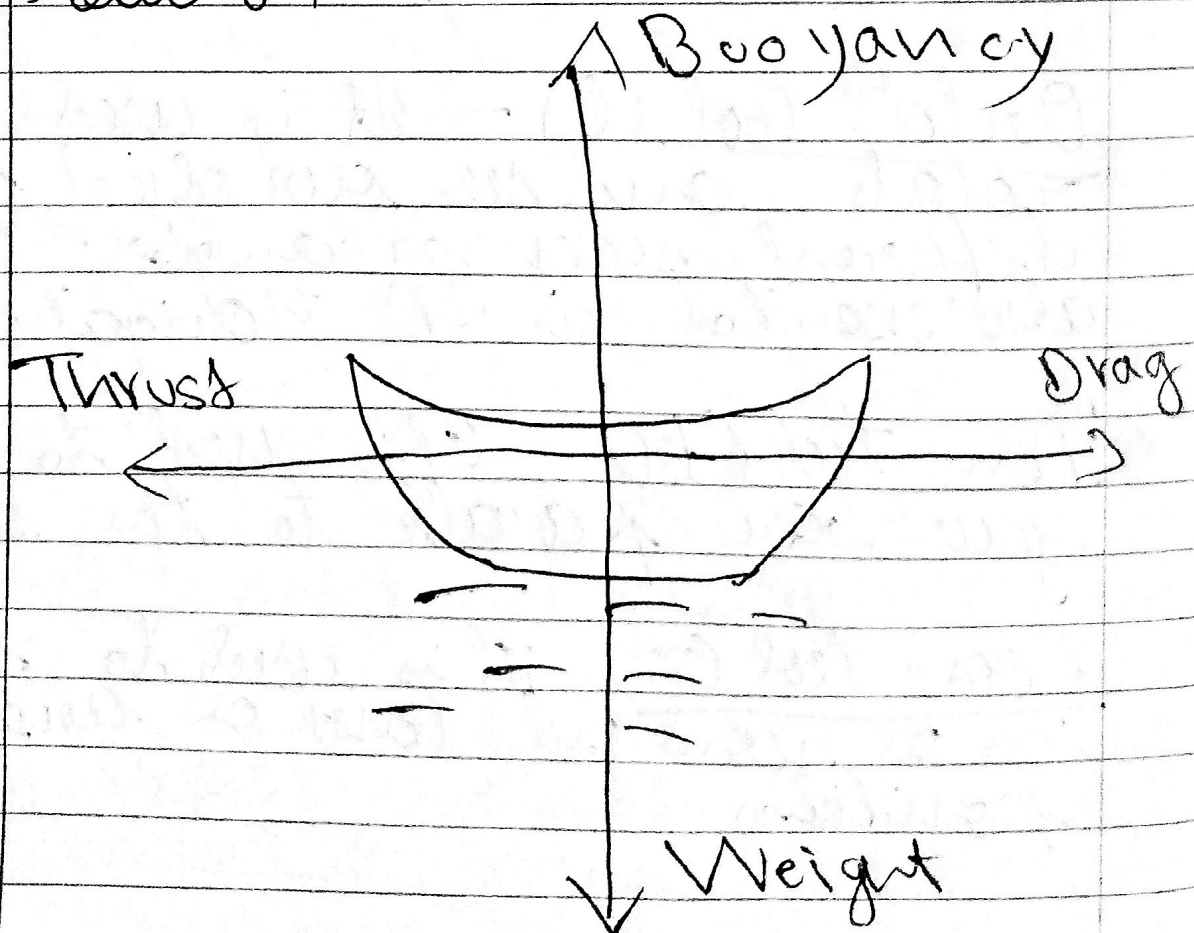
A.BAY

Advanced tool and features

3D text tool → It is used to add editable 3D text to the model.

21/11/25 Propulsion of water

A boat moves forward by applying Newton's third law of motion which states that for every action there is an equal and opposite reaction.



Ohm's law

It is defined as at a constant temp.

$$V \propto I$$

Voltage $\propto$ Current

$$V = RI$$

Resistance $\rightarrow$ Ampere

$R = \text{Resistance } (\Omega) \text{ or (ohm)}$

$I = \text{ampere / Current (A)}$

A current of 5A flowing through a circuit in which a battery of 9V is attached. What is the Resistance in the circuit.

$$\begin{aligned} \because V &= RI \\ \therefore R &= \frac{V}{I} = R = \frac{9}{5} = 1.8 \Omega \end{aligned}$$

Resistance : $\frac{\rho l}{A}$

Resistivity

Length of the wire

Cross section of wire